CLASS X · PRACTICE EXAMINATION

INFORMATION & COMMUNICATION TECHNOLOGY
MODEL QUESTION PAPER 1

Final I pattern · Section A (40) + Section B (60)

Code: BCSE/ICT/MODEL-1 Time: 2 hours 30 minutes

Total: 100 marks Pages: see footer

Instructions. (1) Section A is compulsory: Question 1 carries 40 marks. (2) In Section B answer **ALL** questions (Q2 to Q7, 10 marks each). (3) Write the answers in the spaces given; write program code neatly with correct indentation. (4) Circle only **ONE** letter in the multiple-choice questions. (5) This is a practice paper made for revision; it is not an official question paper.

SECTION A [40 MARKS]

ANSWER **ALL** QUESTIONS

Question 1 **[40]**

a) **Directions: For each question there are four alternatives A, B, C and D. Choose the correct alternative and circle it neatly. Each question carries 2 marks. If more than one letter is circled, NO score will be awarded.** **[30]**

i. Which of the following is an example of Software as a Service (SaaS)?

A Google Docs
B Amazon EC2 virtual server
C Google App Engine
D A school’s own data centre

ii. A school keeps student records on a cloud that only the school can use, hosted in its own building. This is a

A public cloud.
B hybrid cloud.
C private cloud.
D community cloud.

iii. A photograph is shared under the Creative Commons licence **CC BY-ND**. You are allowed to

A change the photograph and claim it as your own.
B sell edited copies without credit.
C share the photograph unchanged and give credit to the creator.
D use it only if you pay the creator.

iv. Which is the correct APA 7 in-text citation for a **direct quotation** from page 14 of a 2022 work by Dorji?

A (2022, Dorji, p.14)
B (Dorji 2022 14)
C (Dorji, 2022, p. 14)
D Dorji (p. 14, 2022)

v. Study the spreadsheet. What is the result of `=SUMIF(B2:B7,"B",C2:C7)`?

	A	B	C
1	**Name**	**Section**	**Marks**
2	Pema	A	72
3	Sonam	B	58
4	Dechen	A	91
5	Karma	B	45
6	Tashi	A	66
7	Yangchen	B	80

A 209
B 183
C 133
D 3

vi. Which chart is the most suitable to show the **share** of each subject in a student's total marks?

A Histogram
B Line chart
C Scatter plot
D Pie chart

vii. What is the output of the program?

```
x = 7
y = 2
print(x // y, x % y)
```

A 3.5 1
B 3 1
C 3 3.5
D 4 1

viii. What is the output of the program?

```
for i in range(2, 10, 3):
    print(i, end=" ")
```

A 2 3 4 5 6 7 8 9
B 2 5 8 11
C 3 6 9
D 2 5 8

ix. What is printed by the program?

```
n = 1
while n < 20:
    n = n * 2
    if n > 10:
        break
print(n)
```

A 8
B 10
C 16
D 32

x. What is the output of the program?

```
s = "Thimphu"
print(s[1:4], s[-1])
```

A Thi u
B hi u
C him p
D him u

xi. What is the output of the program?

```
marks = [40, 55, 70]
marks.append(85)
marks.pop(0)
print(len(marks), marks[0])
```

A 3 55
B 4 40
C 3 40
D 4 55

xii. What is the output of the program?

```
fruit = {"apple": 30, "pear": 45}
fruit["pear"] = 50
fruit["plum"] = 20
print(len(fruit), fruit["pear"])
```

A 2 45
B 3 50
C 3 45
D 2 50

xiii. What is the output of the program?

```
def power(b, e=2):
    return b ** e
print(power(3), power(2, 3))
```

A 9 8
B 9 6
C 6 8
D 3 3

xiv. What does the following print?

```
a = 5
b = 10
print(a > 3 and not b < 8)
```

A False
B True
C Error
D None

xv. What is the output of the program?

```
def f(n):
    if n == 0:
        return 1
    return n * f(n - 1)
print(f(4))
```

A 24
B 12
C 10
D 120

b) Fill in the blanks with suitable word/s. **[5]**

i.	_____ as a Service (IaaS) gives users virtual servers, storage and networks on rent.	
ii.	In an APA reference list the entries are arranged in _____ order of the authors' surnames.	
iii.	The built-in Python function that returns the number of items in a list is _____.	
iv.	A _____ is an unordered collection of unique items written inside curly brackets.	
v.	The keyword used to skip the rest of the current loop turn and go to the next turn is _____.	

c) Write **TRUE** or **FALSE** for each statement in the 'Answer' column. **[5]**

i.	Data saved in the cloud can be opened from any device that has an internet connection.		
ii.	The items of a tuple can be changed after the tuple is created.		
iii.	Using a protected photograph without permission or credit when its licence does not allow it is copyright infringement.		
iv.	In a flowchart, a parallelogram is used for input and output.		
v.	The Excel formula =AVERAGE(B2:B7) only adds the values and does not divide.		

SECTION B [60 MARKS]

ANSWER **ALL** QUESTIONS

Question 2 **[10]**

a) Name the **three** cloud service models and give one example of each. **[3]**

b) Explain **two** ways in which a class can benefit from using Google Docs or Google Slides for a group project. **[2]**

c) The sheet below records the sales of a school shop. Write the Excel formula or function for each. **[4]**

	A	B	C	D	E
1	**Code**	**Item**	**Price (Nu)**	**Qty sold**	**Amount (Nu)**
2	101	Notebook	30	12	
3	102	Pen	10	40	
4	103	Ruler	15	8	
5	104	Bag	450	2	
6		Total			

(i) the amount in **E2** (price × quantity sold)
(ii) the total of the Amount column in **E6**
(iii) the price of the item with code 103, using VLOOKUP
(iv) the largest quantity sold

d) Which chart would you choose to show the **amount** earned from each item, and why? **[1]**

Question 3 **[10]**

a) Define **plagiarism**. Write **two** ways to avoid it. **[3]**

b) What do the Creative Commons symbols **BY** and **NC** mean? **[2]**

c) Write the Python **data type** of each value: (i) `17` (ii) `3.14` (iii) `"Bhutan"` (iv) `False` **[2]**

d) Write the output of the program and show how the value of `total` changes. **[3]**

```
total = 0
for n in range(1, 8):
    if n % 3 == 0:
        continue
    total += n
print("Total:", total)
```

Question 4 **[10]**

a) Write the output of each `print` statement. **[4]**

```
word = "Bhutan Today"
print(word.upper())
print(word[0:6])
print(len(word))
print(word.replace("Today", "Now"))
```

b) Write the output of the program. **[3]**

```
colours = ["red", "green", "blue"]
colours.insert(1, "yellow")
colours.remove("blue")
colours.sort()
print(colours)
```

c) Write the output of the first two `print` statements. Then say what happens when `t[1] = 25` is executed for the tuple `t = (10, 20, 30)`. **[3]**

```
a = {1, 2, 3, 4}
b = {3, 4, 5}
print(a & b)
print(a | b)
```

Question 5 **[10]**

a) Differentiate between a **parameter** and an **argument**. **[2]**

b) Write a function `grade(marks)` that **returns** "Distinction" if marks are 75 or more, "Pass" if marks are 40 to 74, and "Fail" otherwise. **[4]**

c) Trace the recursive function and write its output. **[3]**

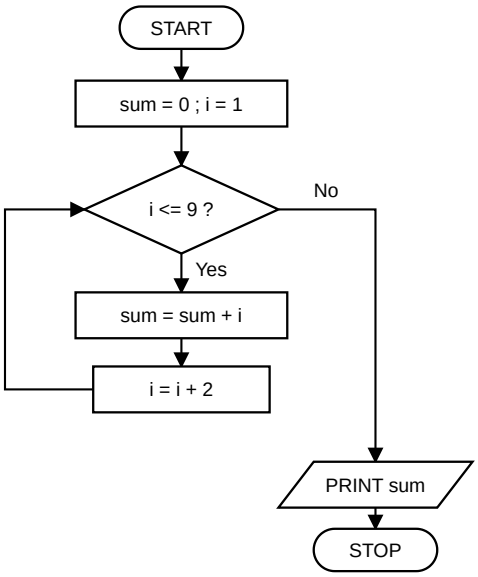
```
def total(n):
    if n == 1:
        return 1
    return n + total(n - 1)
print(total(4))
```

d) Why must every recursive function have a **base case**? **[1]**

Question 6 **[10]**

a) Name the flowchart symbol used for (i) start or stop, (ii) a decision, (iii) input or output, (iv) a process. **[2]**

b) Study the flowchart. What is printed at the end? **[2]**



c) Write a Python program for the flowchart using a `while` loop. **[3]**

d) Give the result (True or False) of: (i) `8 > 5 and 3 > 7` (ii) `not (4 == 4) or 2 < 3` (iii) `10 % 3 == 1` **[3]**

Question 7 **[10]**

A school library charges a late fine of **Nu 2 per day** for every day a book is kept for **more than 14 days**. Write a Python program that:

(a) asks for the number of books borrowed and keeps asking until the number is from 1 to 5 [3]

(b) for each book, asks for the days kept, calculates the fine and stores the fines in a list [4]

(c) prints the fine for each book, then the total fine, or "No fine to pay" if the total is 0 [3]

Rough Work

CLASS X · PRACTICE EXAMINATION

INFORMATION & COMMUNICATION TECHNOLOGY
MODEL QUESTION PAPER 2

Final I pattern · Section A (40) + Section B (60)

Code: BCSE/ICT/MODEL-2 Time: 2 hours 30 minutes

Total: 100 marks Pages: see footer

Instructions. (1) Section A is compulsory: Question 1 carries 40 marks. (2) In Section B answer **ALL** questions (Q2 to Q7, 10 marks each). (3) Write the answers in the spaces given; write program code neatly with correct indentation. (4) Circle only **ONE** letter in the multiple-choice questions. (5) This is a practice paper made for revision; it is not an official question paper.

SECTION A [40 MARKS]

ANSWER **ALL** QUESTIONS

Question 1 **[40]**

a) **Directions: For each question there are four alternatives A, B, C and D. Choose the correct alternative and circle it neatly. Each question carries 2 marks. If more than one letter is circled, NO score will be awarded.** **[30]**

i. A cloud that combines a private cloud with a public cloud is called a

A local cloud.
B personal cloud.
C hybrid cloud.
D closed cloud.

ii. Which Google Workspace tool is the best for collecting answers to a class survey online?

A Google Slides
B Google Meet
C Google Calendar
D Google Forms

iii. Which Creative Commons licence lets others change and share your work, but only if the new work uses the **same licence**?

A CC BY-ND
B CC BY-SA
C CC BY-NC
D CC0

iv. Which is the correct APA 7 reference for a book? (Italics are not shown.)

A Tenzin Wangchuk, 2020, Rivers Of The Himalaya, Royal Press
B Wangchuk, T. (2020). Rivers of the Himalaya. Royal Press.
C (2020) Wangchuk T. Rivers of the Himalaya: Royal Press
D Rivers of the Himalaya. Wangchuk, T. Royal Press. 2020.

v. Study the sheet. What is the value of `=AVERAGE(B2:B5)`?

	A	B
1	**Student**	**Marks**
2	Dema	40
3	Kinley	60
4	Lhamo	80
5	Jigme	20

A 40
B 60
C 50
D 200

vi. Which chart is the best to show how one student's marks **changed over five exams**?

A Pie chart
B Table only
C Column chart of one exam
D Line chart

vii. What is the value printed by this statement?

```
print(2 ** 3 + 10 % 4)
```

A 8
B 9
C 10
D 14

viii. Which output does this program produce?

```
for i in range(3):
    for j in range(2):
        print("*", end="")
    print()
```

A One row of six stars
B Two rows, each with three stars
C Three rows, each with two stars
D Six rows, each with one star

ix. What is the output of the program?

```
count = 0
while count < 3:
    print(count, end=" ")
    count += 1
```

A 0 1 2
B 1 2 3
C 0 1 2 3
D 3

x. What is the output of the program?

```
name = "karma dorji"
print(name.title())
```

A karma dorji
B Karma Dorji
C KARMA DORJI
D Karma dorji

xi. What is the output of the program?

```
nums = [5, 10, 15, 20, 25]
print(nums[-3:])
```

A [5, 10, 15]
B [20, 25]
C [10, 15, 20]
D [15, 20, 25]

xii. What is the output of the program?

```
s = {1, 2, 2, 3, 3, 3}
print(len(s))
```

A 6
B 3
C 5
D 2

xiii. What is the output of the program?

```
d = {"x": 10, "y": 20}
print(d.get("z", 0))
```

A 0
B None
C Error
D 20

xiv. What is the output of the program?

```
def sq(n):
    return n * n
print(sq(sq(2)))
```

A 16
B 8
C 4
D 32

xv. Which loop correctly draws a square with the turtle `t`?

A for i in range(4): t.forward(100); t.left(90)
B for i in range(3): t.forward(100); t.left(90)
C for i in range(4): t.forward(100); t.left(60)
D for i in range(5): t.forward(100); t.left(72)

b) Fill in the blanks with suitable word/s. **[5]**

i.	In a _____ cloud, resources are owned by a provider and shared among many users over the internet.	
ii.	The Excel function that counts the cells that meet a given condition is _____.	
iii.	The Python keyword used to define a function is _____.	
iv.	The _____ statement is used to leave a loop immediately.	
v.	A _____ in Python stores data as key–value pairs.	

c) Write **TRUE** or **FALSE** for each statement in the 'Answer' column. **[5]**

i.	Google Forms can be used to collect responses online.		
ii.	A work in the public domain can be used freely without permission.		
iii.	In Python the index of the first item of a list is 1.		
iv.	A set can store duplicate values.		
v.	A recursive function must have a base case.		

SECTION B [60 MARKS]

ANSWER **ALL** QUESTIONS

Question 2 **[10]**

a) Differentiate between a **public cloud** and a **private cloud** on **two** points. **[3]**

b) State **one** risk of storing data in the cloud and one way of reducing it. **[2]**

c) Study the sheet. Total = English + Maths. The result is "Pass" when the total is 100 or more, otherwise "Fail". **[5]**

	A	B	C	D	E
1	**Name**	**English**	**Maths**	**Total**	**Result**
2	Dema	62	70		
3	Kinley	35	48		
4	Lhamo	80	91		
5	Jigme	55	40		

Write the Excel formula for (i) **D2**; (ii) **E2** using IF; (iii) the highest English mark; (iv) the number of students who passed (use the Result column).

Question 3 **[10]**

a) Write the APA 7 reference for this book: author **Karma Phuntsho**; year **2013**; title **The History of Bhutan**; publisher **Random House India**. **[3]**

b) Explain **two** responsibilities of a good digital citizen. **[2]**

c) Write the output of the program. **[3]**

```
for i in range(1, 4):
    for j in range(i):
        print(i, end=" ")
    print()
```

d) State the difference between the `break` and `continue` statements. **[2]**

Question 4 **[10]**

a) Write the output of the program. **[4]**

```
stock = {"rice": 20, "dal": 12}
stock["oil"] = 8
stock["rice"] -= 5
del stock["dal"]
print(stock)
print(len(stock))
```

b) Write the output of each `print` statement. **[3]**

```
text = "python programming"
print(text.count("g"))
print(text.find("pro"))
print(text[::-1])
```

c) Write the output of the program. **[3]**

```
A = {2, 4, 6, 8}
B = {4, 8, 10}
print(A | B)
print(A & B)
print(A - B)
```

Question 5 **[10]**

a) Write a function `interest(p, r, t)` that **returns** the simple interest (p × r × t ÷ 100). Then show a statement that calls it to print the interest on Nu 5000 at 8% for 2 years. **[3]**

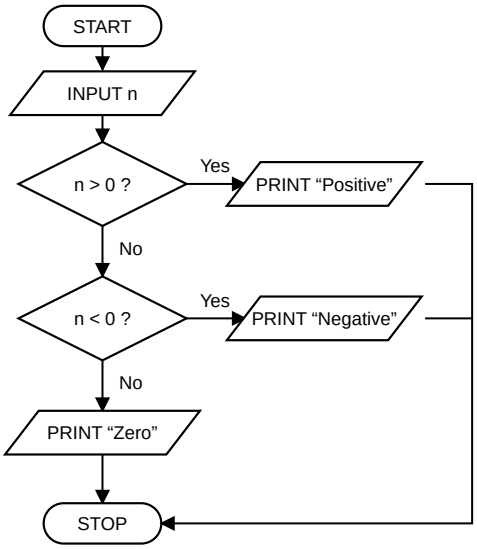
b) Write the output of the program. **[3]**

```
def mystery(a, b=3):
    return a * b + 1
print(mystery(2))
print(mystery(2, 5))
print(mystery(b=4, a=1))
```

c) Write a **recursive** function `sum_to(n)` that returns the sum of the first n natural numbers (1 + 2 + … + n). Show the call for n = 5 and its output. **[4]**

Question 6 **[10]**

a) Study the flowchart. (i) What is printed when $n = 7, -3$ and 0? (ii) Write the Python program for the flowchart using `if – elif – else`. **[6]**



b) Complete the program so that the turtle draws a regular pentagon with sides of 80 steps. **[2]**

```
import turtle
t = turtle.Turtle()
for i in range(___):
    t.forward(___)
    t.right(___)
```

c) Give the result of: (i) `not True or False` (ii) `(5 > 2) and (4 < 1)` (iii) `7 % 2 == 1 or 1 > 2` **[2]**

Question 7 **[10]**

A school canteen keeps its menu in a **dictionary**: momo Nu 60, tea Nu 15, noodles Nu 80, juice Nu 30. Write a Python program that:

(a) stores the menu in a dictionary [2]

(b) repeatedly asks for the item (type *done* to stop); if the item is in the menu, asks for the quantity (must be above 0) and adds the cost to the total, otherwise prints "Item not on the menu" [5]

(c) gives a **10% discount** if the total is more than Nu 200 [1]

(d) prints the total, the discount and the amount to pay [2]

Rough Work

ANSWER KEYS

ICT Model Paper 1 and Model Paper 2

ANSWER KEY — MODEL PAPER 1

BCSE/ICT/MODEL-1 · for self-checking

Section A — Q1(a) Multiple choice (2 marks each)

No.	Answer	Reason
i.	**A**	Google Docs is software used through a browser: SaaS. EC2 is IaaS and App Engine is PaaS.
ii.	**C**	A cloud used by one organisation only is a private cloud.
iii.	**C**	BY = give credit; ND = no derivatives, so it can be shared but not changed.
iv.	**C**	APA 7 direct quote: (Author, year, p. page).
v.	**B**	Section B marks are 58 + 45 + 80 = 183.
vi.	**D**	A pie chart shows parts of a whole.
vii.	**B**	7 // 2 = 3 (floor division) and 7 % 2 = 1 (remainder).
viii.	**D**	range(2, 10, 3) gives 2, 5, 8.
ix.	**C**	n becomes 2, 4, 8, 16; 16 > 10 so the loop breaks and 16 is printed.
x.	**D**	s[1:4] = “him” (indexes 1, 2, 3); s[-1] = “u”.
xi.	**A**	After append the list has 4 items; pop(0) removes 40, leaving [55, 70, 85].
xii.	**B**	pear is changed to 50 and plum is added, so there are 3 keys.
xiii.	**A**	power(3) uses e = 2 → 9; power(2, 3) = 2^3 = 8.
xiv.	**B**	a > 3 is True; b < 8 is False, so not False is True; True and True = True.
xv.	**A**	f(4) = 4 × 3 × 2 × 1 × f(0) = 24.

Q1(b) Fill in the blanks · Q1(c) True/False (1 mark each)

	(b)	(c)
i.	**Infrastructure**	**TRUE**
ii.	**alphabetical**	**FALSE**
iii.	**len()**	**TRUE**
iv.	**set**	**TRUE**
v.	**continue**	**FALSE**

Section B (10 marks per question; all compulsory) — columns: part, marks, model answer

Question 2

a)	3	IaaS – Infrastructure as a Service, e.g. Amazon EC2 / virtual servers [1]; PaaS – Platform as a Service, e.g. Google App Engine [1]; SaaS – Software as a Service, e.g. Google Docs, Gmail [1]. (Accept any correct example.)
b)	2	Any two, 1 mark each: several students can edit the same file **at the same time** from different places; changes are **saved automatically** and the version history lets them go back; **comments and suggestions** let members give feedback; the file can be **shared by link** with view/comment/edit rights; no need to email copies.
c)	4	(i) `=C2*D2` (= 360) [1]; (ii) `=SUM(E2:E5)` (= 1780) [1]; (iii) `=VLOOKUP(103,A2:C5,3,FALSE)` (= 15) [1]; (iv) `=MAX(D2:D5)` (= 40) [1]. Amounts: 360, 400, 120, 900.
d)	1	Column (bar) chart [½]: it lets the amounts of the four items be compared easily [½]. (A pie chart is also accepted if the reason is given.)

Question 3

a)	3	Plagiarism is using another person's words, ideas or work and presenting it as your own without giving credit [1]. Avoid it by: writing in your own words and **citing the source** [1]; putting quotation marks around copied words and adding a reference list [1]; asking permission / using work with a suitable licence.
b)	2	BY (Attribution) – users must give credit to the creator [1]. NC (Non-Commercial) – the work may not be used for profit [1].
c)	2	(i) int (ii) float (iii) str (string) (iv) bool (Boolean) — ½ mark each.
d)	3	n takes 1, 2, 3, 4, 5, 6, 7. Multiples of 3 (3 and 6) are skipped by `continue`. total = 1 → 3 → 7 → 12 → 19 [2]. Output: `Total: 19` [1]

Question 4

a)	4	Output [1 each]:
		`BHUTAN TODAY`
		`Bhutan`
		`12`
		`Bhutan Now`

b)	3	insert → ['red', 'yellow', 'green', 'blue']; remove → ['red', 'yellow', 'green']; sort → alphabetical [2]. Output: `['green', 'red', 'yellow']` [1]

c)	3	a & b is the intersection and a | b is the union:
		`{3, 4}`
		`{1, 2, 3, 4, 5}`
		[2] `t[1] = 25` gives a **TypeError** because a tuple is immutable (its items cannot be changed) [1].

Question 5

a)	2	A parameter is the variable named in the function definition (`def area(l, w)`) [1]; an argument is the actual value passed when the function is called (`area(5, 3)`) [1].

b)	4	`def grade(marks):`
		`  if marks >= 75:`
		`  return "Distinction"`
		`  elif marks >= 40:`
		`  return "Pass"`
		`  else:`
		`  return "Fail"`
		def line and colon [1]; first condition and return [1]; elif for 40–74 [1]; else Fail [1].

c)	3	total(4) = 4 + total(3) = 4 + (3 + total(2)) = 4 + 3 + (2 + total(1)) = 4 + 3 + 2 + 1 [2]. Output: **10** [1].
d)	1	The base case is the condition that stops the function from calling itself; without it the recursion never ends and causes a RecursionError (stack overflow) [1].

Question 6

a)	2	(i) terminator / oval (rounded rectangle); (ii) diamond; (iii) parallelogram; (iv) rectangle — ½ mark each.
b)	2	sum = 0, i = 1: i = 1, 3, 5, 7, 9 are added (the loop stops when i = 11): 1 + 3 + 5 + 7 + 9 = **25** [2].

c)	3	`total = 0`
		`i = 1`
		`while i <= 9:`
		`  total = total + i`
		`  i = i + 2`
		`print(total)`
		initialise sum and i [1]; `while i <= 9` with the two updates inside [1]; `print(total)` after the loop [1].

d)	3	(i) False (True and False); (ii) True (False or True); (iii) True (10 % 3 = 1) — 1 mark each.

Question 7

	10	

```
n = int(input("Number of books borrowed (1-5): "))
while n < 1 or n > 5:
    print("Invalid. Enter a number from 1 to 5.")
    n = int(input("Number of books borrowed (1-5): "))

fines = []
for i in range(1, n + 1):
    days = int(input("Days kept for book " + str(i) + ": "))
    if days > 14:
        fine = (days - 14) * 2
    else:
        fine = 0
    fines.append(fine)

total = 0
for i in range(len(fines)):
    print("Book", i + 1, "fine: Nu", fines[i])
    total = total + fines[i]

if total == 0:
    print("No fine to pay")
else:
    print("Total fine: Nu", total)
```

Sample run (inputs 7, 3, 10, 20, 15):

```
Number of books borrowed (1-5): Invalid. Enter a number from 1 to 5.
Number of books borrowed (1-5): Days kept for book 1: Days kept for book 2:
Days kept for book 3: Book 1 fine: Nu 0
Book 2 fine: Nu 12
Book 3 fine: Nu 2
Total fine: Nu 14
```

Marking: (a) input and conversion to int [1], while loop with the condition `n < 1 or n > 5` [1], message and re-input [1]. (b) empty list [1], for loop over the books [1], correct fine `(days - 14) * 2` only when days > 14 [1], append to list [1]. (c) loop to print each fine [1], total [1], if/else for “No fine” [1]. Accept any correct program.

ANSWER KEY — MODEL PAPER 2

BCSE/ICT/MODEL-2 · for self-checking

Section A — Q1(a) Multiple choice (2 marks each)

No.	Answer	Reason
i.	**C**	Private + public = hybrid cloud.
ii.	**D**	Google Forms makes online surveys and records the responses in a sheet.
iii.	**B**	SA = ShareAlike: derivative works must carry the same licence.
iv.	**B**	APA 7: Surname, Initial. (Year). Title in sentence case. Publisher.
v.	**C**	(40 + 60 + 80 + 20) ÷ 4 = 50.
vi.	**D**	A line chart shows a trend over time.
vii.	**C**	2 ** 3 = 8 and 10 % 4 = 2, so 8 + 2 = 10.
viii.	**C**	The outer loop runs 3 times; each time the inner loop prints 2 stars, then print() starts a new line.
ix.	**A**	count = 0, 1, 2 are printed; when count becomes 3 the condition is false.
x.	**B**	title() makes the first letter of every word a capital.
xi.	**D**	nums[-3:] takes the last three items.
xii.	**B**	A set keeps only unique values: {1, 2, 3}.
xiii.	**A**	The key “z” is missing, so get() returns the default value 0.
xiv.	**A**	sq(2) = 4, then sq(4) = 16.
xv.	**A**	A square has 4 equal sides and 90° turns.

Q1(b) Fill in the blanks · Q1(c) True/False (1 mark each)

	(b)	(c)
i.	**public**	**TRUE**
ii.	**COUNTIF**	**TRUE**
iii.	**def**	**FALSE**
iv.	**break**	**FALSE**
v.	**dictionary**	**TRUE**

Section B (10 marks per question; all compulsory) — columns: part, marks, model answer

Question 2

a)	3	Any two points [1½ each]: owner – public: a provider (e.g. Google, Amazon); private: one organisation. Users – public: many users share the resources; private: only that organisation. Cost – public: pay for what is used, cheap to start; private: costly to set up and maintain. Security/control – private gives more control; public less.
b)	2	Risk [1]: data can be hacked or leaked / service may be unavailable without internet / provider may lose data. Reduction [1]: use strong passwords and two-step verification / keep a backup copy / set correct sharing permissions.
c)	5	(i) `=B2+C2` [1]; (ii) `=IF(D2>=100,"Pass","Fail")` [2: IF with condition 1, both results 1]; (iii) `=MAX(B2:B5)` [1]; (iv) `=COUNTIF(E2:E5,"Pass")` [1] = 2 (Dema 132 and Lhamo 171 pass; Kinley 83 and Jigme 95 fail).

Question 3

a)	3	Phuntsho, K. (2013). *The history of Bhutan*. Random House India. — surname and initial [1]; year in brackets [1]; title in italics and sentence case, then publisher [1].
b)	2	Any two [1 each]: be respectful and kind online (no cyber-bullying); protect your privacy and passwords; give credit and avoid plagiarism; check facts before sharing; follow copyright and licence rules; do not share others' photos without permission.

c)	3	The outer loop gives i = 1, 2, 3 and the inner loop prints i exactly i times on the same line [2]. Output [1]:
		`1`
		`2 2`
		`3 3 3`

d)	2	`break` ends the whole loop at once and the program goes to the next statement after the loop [1]. `continue` skips the rest of the current turn only and starts the next turn of the loop [1].

Question 4

a)	4	rice becomes 15 (20 − 5), oil is added, dal is deleted [2]. Output [2]:
		`{'rice': 15, 'oil': 8}`
		`2`

b)	3	count("g") = 2 [1]; find("pro") = 7 (the index where "pro" starts in "programming") [1]; reversed text [1]:
		`2`
		`7`
		`gnimmargorp nohtyp`

c)	3	Union, intersection, difference [1 each]:
		`{2, 4, 6, 8, 10}`
		`{8, 4}`
		`{2, 6}`

Question 5

a)	3	`def interest(p, r, t):`
		`  return p * r * t / 100`
		
		`print(interest(5000, 8, 2))`
		def with 3 parameters [1]; correct return expression [1]; correct call/print (800.0) [1].

b)	3	mystery(2) = 2 × 3 + 1 = 7; mystery(2, 5) = 2 × 5 + 1 = 11; mystery(b=4, a=1) = 1 × 4 + 1 = 5. Output (1 mark each):
		`7`
		`11`
		`5`

c)	4	`def sum_to(n):`
		`  if n == 1:`
		`  return 1`
		`  return n + sum_to(n - 1)`
		
		`print(sum_to(5))`
		base case n == 1 [1]; recursive call with n − 1 [1]; adding n to the call [1]; output 15 [1].

Question 6

a)	6	(i) 7 → Positive; −3 → Negative; 0 → Zero [2]. (ii)
		`n = int(input("Enter a number: "))`
		`if n > 0:`
		`  print("Positive")`
		`elif n < 0:`
		`  print("Negative")`
		`else:`
		`  print("Zero")`
		input with int [1]; if n > 0 [1]; elif n < 0 [1]; else Zero [1].

b)	2	range(**5**), forward(**80**), right(**72**) — a pentagon has 5 sides and the exterior angle is 360 ÷ 5 = 72° [½, ½, 1].
c)	2	(i) False; (ii) False; (iii) True. [1 for two correct, 2 for all three]

Question 7

	10	```
		menu = {"momo": 60, "tea": 15, "noodles": 80, "juice": 30}
		total = 0
		
		print("Menu:", list(menu.keys()))
		item = input("Enter item (or done): ").lower()
		while item != "done":
		    if item in menu:
		        qty = int(input("Quantity: "))
		        while qty <= 0:
		            qty = int(input("Quantity must be positive. Quantity: "))
		        total = total + menu[item] * qty
		    else:
		        print("Item not on the menu")
		    item = input("Enter item (or done): ").lower()
		
		discount = 0
		if total > 200:
		    discount = total * 10 / 100
		
		print("Total:", total)
		print("Discount:", discount)
		print("Amount to pay:", total - discount)
		``
		Sample run (inputs momo, 3, pizza, tea, 0, 2, done):
		``
		Menu: ['momo', 'tea', 'noodles', 'juice']
		Enter item (or done): Quantity: Enter item (or done): Item not on the menu
		Enter item (or done): Quantity: Quantity must be positive. Quantity: Enter
		item (or done): Total: 210
		Discount: 21.0
		Amount to pay: 189.0
		``
		Marking: (a) dictionary with 4 correct pairs [2]. (b) while loop until “done” [1], `in menu` test [1], quantity validation [1], cost = price × quantity added to total [1], else message [1]. (c) if total > 200 then 10% [1]. (d) three correct prints [2]. Accept any correct program.

Marking guide: in Section B each bracketed point is worth the mark shown; accept any correct program or answer written in different words. For programs, minor spelling slips in names are not penalised when the logic is correct, but missing colons or wrong indentation can lose the mark for that line. For “write the output” questions the output must be exactly as printed (one line per `print` unless `end` is used).